CLOSED IMMERSIONS CANNOT INCREASE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original lemma 0DRN: sources/stacks-chow/tags.txt, source lines 17228–17228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *If $\mathcal{Z} \hookrightarrow \mathcal{X}$ is a closed immersion of locally Noetherian algebraic stacks, and if $z \in |\mathcal{Z}|$ has image $x \in |\mathcal{X}|$, then $\dim_z(\mathcal{Z}) \leq \dim_x(\mathcal{X})$.*

Proof. Choose a smooth surjective morphism $U \to \mathcal{X}$ whose source is a scheme; the base-changed morphism $V = U \times_{\mathcal{X}} \mathcal{Z} \to \mathcal{Z}$ is then also smooth and surjective, and the projection $V \to U$ is a closed immersion. If $v \in |V|$ maps to $z \in |\mathcal{Z}|$, and if we let u denote the image of v in $|U|$, then clearly $\dim_v(V) \leq \dim_u(U)$, while $\dim_v(V_z) = \dim_u(U_x)$, by Lemma 0DRN. Thus

$$\dim_z(\mathcal{Z}) = \dim_v(V) - \dim_v(V_z) \leq \dim_u(U) - \dim_u(U_x) = \dim_x(\mathcal{X}),$$

as claimed. □

Date: stacks-project:stacks-geometry.tex:local-snapshot-20261012; GNU FDL 1.2 or later.